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Data Article

Dataset of high-frequency water quality and meteorological variables in Buffalo Pound Lake, Saskatchewan, Canada, 2014 – 2021

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Dataset link: A long-term water quality and meteorological data set (2014 – 2021) of a eutrophic prairie lake: Buffalo Pound Lake, Saskatchewan, Canada (Original data)

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ABSTRACT

Lakes can undergo rapid changes that are not captured during traditional, discrete sampling campaigns. Sensor-based data provide opportunities to understand these rapid changes in lakes. Here, we present eight years of sensor-based monitoring data from the open water season in a shallow, polymictic reservoir in southern Saskatchewan, which serves as an important drinking water supply. A monitoring buoy was moored annually at the same site providing sensor data including photosynthetically active radiation (PAR; 0.62 and 0.78 m below surface), pH, dissolved oxygen, turbidity, specific conductivity, phycocyanin and chlorophyll (0.8, or 0.8 and 2.8 m below surface), and temperature (multiple locations in the water column) at high frequency. The buoy was also equipped with a weather station to record air temperature, barometric pressure, PAR, rain, relative humidity, wind direction and wind speed. Data were reviewed for data quality. This long-term dataset can be used to understand thermal variation, chemical and ecological change, and to charac-

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terize the often rapid changes that polymictic lakes undergo, particularly those resulting from seasonal changes and development of cyanobacterial blooms.

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Specifications Table

Subject	Earth & Environmental Sciences
Specific subject area	Environmental monitoring, water quality, freshwater ecology
Type of data	Processed data tables (csv)
Data collection	Data were acquired using a series of sensors (YSI sondes, Nexsens T-node thermistors, LI-COR photovoltaic sensor), affixed to a stationary floating buoy, with a buoy-mounted weather station (Vaisala). Data were collected every 15 min in 2014 and every 10 min from 2015 to 2021 during the open water deployment season. Buoy sensor data processing involved investigating of field-flagged concerns and automated screening for data quality. All known errors—such as sensor or system failures and technician interference—were reviewed, and erroneous measurements were removed to ensure dataset reliability.
Data source location	Buffalo Pound Lake, Saskatchewan, Canada Geolocation Box West Longitude: −105.387343 East Longitude: −105.383540 North Latitude: 50.586310 South Latitude: 50.585611
Data accessibility	See Baulch et al. 2025 [1] Repository name: Federated Research Data Repository Data identification number: DOI 10.20383/103.01100
Related research article	*None.*

1. Value of the Data

- These data support insights into the rapid changes that can occur in polymictic lakes that are observable via high frequency monitoring.
- The data will support comparative work across lakes to help better understand common drivers of change during the open water season across diverse regions and lakes, along with more localized phenomena.
- Thermal data and weather data can also support climatic and hydrological research, and all data will be of benefit to lake modelling efforts.
- This monitoring has informed drinking water treatment, and ongoing analyses can be used to better understand driving factors affecting water quality, and its treatability in a rapidly changing environment.

2. Background

High frequency lake monitoring data allow us to see changes in lakes that may drive their ecology but are not captured during traditional (often weekly to monthly) sampling campaigns [2]. The monitoring buoy and sensor system used in this research has supported new insights into lake ecology [3,4], comparative work across lakes [5] combined with beneficial insights on how changes in source water quality can influence water treatment [3,6]. The data presented here have multiple uses, including supporting work on lake metabolism, biogeochem-

istry, cyanobacterial dynamics, comparative work across ecosystems, as well as work to help water treatment adapt to changing source water quality. Importantly, it helps fill gaps in our understanding of shallow, managed reservoirs, and drivers of water chemistry change in the northern Great Plains, a vast region where issues of water scarcity and water quality concerns are common.

3. Data Description

The dataset includes cleaned, subhourly weather and water quality data from Buffalo Pound Lake, Saskatchewan, Canada, collected during the ice-free season. Data files are organized by year from 2014 to 2021 with measurements including air temperature, relative humidity, wind speed, wind direction, daily rain, photosynthetically active radiation (PAR) above and below water surface (at two depths), thermistors at seven depths, and pH, dissolved oxygen, specific conductivity and phycocyanin at two depths. Turbidity and chlorophyll measurements are available for a single depth (shallow depth; i.e., 0.8 m), through the study period. Additional sensors were deployed in the latter years.

All data files are stored in the BuffaloPoundLake_WQmonitoringbuoy directory as eight CSV files using the naming convention BPBuoyData_[YEAR]_Cleaned.csv, where [YEAR] corresponds to the collection year.

The data files use a wide format where rows represent unique timestamps and columns represent measured environmental variables or data quality flags.

- Of 68 total columns, 33 represent types of unique environmental observations, while the remainder indicate timestamp, battery level and data quality flags.
- The measurement and data quality flag columns typically alternate within the data files. This structure enables independent data quality assessment for each environmental variable. This is important given the system design, where individual sensor issues often do not affect other measurements.
- Variable names follow a consistent convention indicating unit of measure and relative depth positions (e.g., "Shallow" and "Deep" designators, independent thermistors are numbered from Temp00 to Temp04 for shallowest to deepest, respectively), with actual depths specified in the README file and in Table 1.

Additional (supporting) documentation includes:

- README.txt – Plain text file describes file organization, summarizes annual data collection periods, defines missing data codes and describes variables (see also Table 1 for variable descriptions).
- data_flags.csv – CSV file defines data quality flags and codes used in the dataset (see also Table 2).

Data are collected continuously through the deployment, with the exception of some data gaps associated with issues and events including sensor failure or maintenance. Periodic data quality issues are also seen, associated with fouling and other issues (e.g., see Table 2). In addition to a review of field-based notes that highlighted period of fouling and/or instrument damage, an automated screening approach was used to systematically screen all data. This was done by applying flags specifying missing data, values outside sensor operating ranges, values that exceed expected environmental ranges (based on historic data), or values which exhibit unusual patterns such as repeated values or erratic rate changes which can be indicative of equipment failure or fouling. All flagged data underwent manual review to assess data quality and environmental plausibility, using complementary data (field-based measurements or measurements of the Buffalo Pound drinking water treatment plant) where possible (See section: Experimental design, materials and methods for further details). Following quality control the final dataset spans 8 years, 142,760 records (i.e., unique time stamps) and tens of thousands of observations

Table 1
Description of variables (as column names) in the dataset. Each environmental variable column has an associated quality flag column (denoted by '_Flag' suffix). Flag definitions and criteria are provided in the accompanying 'data_flags.csv' file.

Column name	Description
DateTime	Date and time of measurement [YYYY-MM-DD HH:MM:SS]
Bat_V	Battery voltage [V]
BarometricPress_kPa	Barometric pressure [kPa]
RelativeHum_%	Relative humidity [%]
WindDir_Degree	Wind direction [degrees]
WindSp_km/h	Wind speed [km h^{-1}]
DailyRain_mm	Rainfall [mm]
AirTemp_C	Air temperature [degrees C]
PARAir_umol/s/m^2	Photosynthetically active radiation in air [µmol s^{-1} m^{-2}]
PARW1_umol/s/m^2	Photosynthetically active radiation in water at 0.62 m depth [µmol s^{-1} m^{-2}]
PARW2_umol/s/m^2	Photosynthetically active radiation in water at 0.78 m depth [µmol s^{-1} m^{-2}]
TempShallow_C	Water temperature at 0.8 m depth [degrees C]
pHShallow	pH at 0.8 m depth [pH units]
pHmVShallow_mV	pH in millivolts at 0.8 m depth [mV]
ODOSatShallow_%	Dissolved oxygen saturation at 0.8 m depth [%]
ODOShallow_mg/L	Dissolved oxygen concentration at 0.8 m depth [mg/L]
SpCondShallow_uS/cm	Specific conductivity at 0.8 m depth [µS cm^{-1}]. 2017 data include corrections applied during quality assurance procedures to rectify calibration errors.
SpCondShallowRaw_uS/cm	Raw specific conductivity at 0.8 m depth [µS cm^{-1}]
TurbShallow_NTU+	Turbidity at 0.8 m depth [NTU]
ChlRFUShallow_RFU	Chlorophyll relative fluorescence units at 0.8 m depth [RFU]
BGPCShallowRFU_RFU	Blue-green phycocyanin relative fluorescence units at 0.8 m depth [RFU]
TempDeep_C	Water temperature at 2.8 m depth [degrees C]
pHDeep	pH at 2.8 m depth [pH units]
pHmVDeep_mV	pH in millivolts at 2.8 m depth [mV]
ODODeep_mg/L	Dissolved oxygen concentration at 2.8 m depth [mg L^{-1}]
ODOSatDeep_%	Dissolved oxygen saturation at 2.8 m depth [%]
SpCondDeep_uS/cm	Specific conductivity at 2.8 m depth [µS cm^{-1}]. 2017 and 2018 data include corrections applied during quality assurance procedures to rectify calibration errors.
SpCondDeepRaw_uS/cm	Raw specific conductivity at 2.8 m depth [µS cm^{-1}]
ChlorophyllRFUDeep_RFU	Chlorophyll at 2.8 m depth [RFU]
BGPCDeepRFU_RFU	Blue-green phycocyanin at 2.8 m depth [RFU]
Temp00_C	Temperature at 0.45 m depth [degrees C]
Temp01_C	Temperature at 0.77 m depth [degrees C]
Temp02_C	Temperature at 1.23 m depth [degrees C]
Temp03_C	Temperature at 2.18 m depth [degrees C]
Temp04_C	Temperature at 3.18 m depth [degrees C]

per parameter (Table 3). During this period environmental conditions spanned a broad range, including >2 pH units, 20 °C, and 2-fold variation in conductivity. Pigment-related indicators demonstrated variation ranging from clearwater conditions to extreme blooms (Table 4, Fig. 1).

4. Experimental Design, Materials and Methods

4.1. Site description

Buffalo Pound Lake is a eutrophic, polymictic reservoir in south-central Saskatchewan, which is a semi-arid region with highly agricultural land use, and frequent issues associated with poor water quality and water scarcity. The lake is long, and narrow (29 km long and <1 km wide), and subject to long-periods of ice cover (between 124 and 184 days of ice cover per year), with ice-on typically in November or December and ice-off in April or May) [7]. This multi-use reservoir was impounded first in 1939 [8], with subsequent work to increase water levels; flow to the

Table 2
Data flags and associated definitions indicating missing, potentially suspect, or revised data.

Flag	Definition
M	Indicates data is missing, sensor was removed during deployment due to damage or failure or values were not recorded or logged due to internal errors
C	Indicates data that was deemed faulty or unrealistic.
A2	Photosynthetically active radiation (PAR) measurement below factory range (2 - 2,000 µmol s^{-1} m^{-2}) during periods of low light. Value adjusted to 0.
A3	Inverted PAR readings may represent environmental conditions such as algae blooms, sediment plumes, localized obstructions, shadows from boats, differential sensor fouling, or measurement error
A4	In 2015, underwater PAR sensors were positioned on either side of the buoy with intermittent shading occurring across the study season. We recommend using the highest value of the two sensors, but not using them in a comparative way to understand light attenuation between the two sensor depths due to shading effects associated with sensor orientation in this year.
A5	Daily rainfall measurement indicates a sudden, localized rainfall event undetected by nearby Moose Jaw CS Saskatchewan weather station (https://climate.weather.gc.ca/historical_data/search_historic_data_e.html)
A6	Corrected Specific Conductivity measurements in response to issues surrounding calibration solutions, i) shallow (0.8 m) and deep (2.8 m) sensors during 2017 AND ii) deep (2.8 m) sensors during 2018.
B6	Raw Specific Conductivity measured show stepwise changes in response to issues surrounding calibration solutions, i) shallow (0.8 m) and deep (2.8 m) sensors during 2017 and ii) deep (2.8 m) sensors during 2018.
B7	Chlorophyll and/or phycocyanin and/or Turbidity measurements showed a spike in concentration combined with a noticeable decrease following site visit.

Table 3
Valid observation counts per year for select environmental variables, 2014 – 2021. (Zero counts reflect long-term sensor failure). See Table 1 for variable descriptions (i.e., as column names).

Column name	2014	2015	2016	2017	2018	2019	2020	2021
AirTemp_C	7,993	20,419	18,268	16,654	17,053	17,959	16,836	18,279
RelativeHum_%	7,997	20,419	18,788	16,689	17,053	17,967	16,836	18,282
WindSp_km/h	7,997	20,419	14,065	16,714	17,053	17,967	16,833	18,282
PARAir_umol/s/m^2	8,002	20,425	18,757	22,004	15,051	18,026	16,836	17,220
PARW1_umol/s/m^2	6,743	20,425	18,975	22,022	17,053	18,026	16,836	1,376
PARW2_umol/s/m^2	6,615	20,425	18,750	22,022	17,053	18,026	16,836	17,258
TempDeep_C	8,002	20,339	18,970	22,015	17,719	1,163	16,834	18,285
TempShallow_C	8,002	17,288	18,717	22,014	17,723	17,862	15,984	18,260
pHDeep	8,002	20,339	18,955	12,335	17,721	1,163	16,834	0
pHShallow	8,002	17,288	18,719	22,020	17,724	17,862	15,987	18,261
ODOSatDeep_%	8,002	20,339	18,743	21,916	17,665	501	16,780	18,277
ODOSatShallow_%	8,002	7,872	18,708	21,979	17,724	17,862	15,984	17,884
SpCondDeep_uS/cm	8,002	20,334	18,942	21,637	17,671	1,162	16,825	18,282
SpCondShallow_uS/cm	8,002	17,288	18,353	21,992	17,718	17,857	15,966	18,260
Temp00_C	0	20,425	18,965	22,007	17,042	18,026	16,824	17,607
Temp01_C	8,002	20,425	18,965	22,000	17,045	18,026	16,822	18,036
Temp02_C	8,001	20,425	18,971	22,015	17,047	18,026	16,833	18,053
Temp03_C	8,001	20,425	18,974	22,022	17,048	18,025	16,829	18,067
Temp04_C	0	20,425	18,948	22,016	17,049	18,026	16,830	17,839

reservoir was later increased following construction of an upstream reservoir (Lake Diefenbaker) which permitted water diversions into the Buffalo Pound system [8,9].

The lake serves as a water supply to approximately one-quarter of the province of Saskatchewan's population (including the cities of Regina and Moose Jaw), as well as providing industrial water. Buffalo Pound Lake also supports numerous cultural and recreational uses, including fishing. The shallow nature of the lake (average depth of 3.0 m and a maximum depth of <6 m) means that it can undergo transient stratification [6].

Table 4
Summary statistics for select environmental variables across the complete dataset (2014 – 2021 deployments). See Table 1 for variable descriptions (i.e., as column names).

Column name	Count	Mean	Min	SD	25 %	50 %	75 %	Max
DateTime	142,760	–	–	–	–	–	–	–
AirTemp_C	133,461	17	−4.3	5.7	13.4	17.1	20.8	39.3
RelativeHum_%	134,031	64.9	11	17.1	52	67	79	96
WindSp_km/h	129,330	16.1	0.2	10	8.4	14.2	21.7	82.4
PARAir_umol/s/m^2	136,321	311.5	0	411.2	7	91	519	2538
PARW1_umol/s/m^2	121,456	99.4	0	185.8	0	21	109	1759
PARW2_umol/s/m^2	136,985	64.4	0	133.1	0	10	64	1534
TempDeep_C	123,327	18.6	5.8	3.5	16.2	19.2	21.4	24.6
TempShallow_C	135,850	18.9	5.6	3.4	16.8	19.4	21.7	26.5
pHDeep	95,349	8.6	7.6	0.3	8.4	8.6	8.7	9.7
pHShallow	135,863	8.5	7.5	0.3	8.3	8.5	8.7	9.5
ODOSatDeep_%	122,223	89.9	3.1	18.6	80.7	91.4	100.3	169
ODOSatShallow_%	126,015	101.4	31.5	22.1	87.9	96.9	110	280
SpCondDeep_uS/cm	122,855	697.1	418	183.1	498	742	856	995
SpCondShallow_uS/cm	135,436	669.1	411	179.6	492	621	835	991
Temp00_C	130,896	18.9	5.8	3.6	16.4	19.3	21.8	28.6
Temp01_C	139,321	18.8	5.7	3.5	16.5	19.3	21.7	28.2
Temp02_C	139,371	18.7	5.7	3.5	16.4	19.2	21.6	26.8
Temp03_C	139,391	18.6	5.7	3.5	16.4	19.1	21.5	25.2
Temp04_C	131,133	18.3	5.7	3.4	16	18.8	21	24.5

Water in the lake comes from an upstream reservoir (Lake Diefenbaker), in addition to the local catchment (and precipitation to the lake surface as well potential groundwater inputs). The high degree of hydroclimatic variation in the region (combined with snowmelt-driven hydrology and thresholding hydrological processes) contribute to high variation in flows and flow source [10,11]. Long-term monitoring of flows and modelling show an important influence of climate, flow source and water management on water quality [10,12–14]. Paleolimnological data show a slow degradation of water quality over time [15], with long-term monitoring also showing transient increases in odor and other parameters of concern [10,12,16].

High frequency data show rapid temporal changes can occur in the lake. Transient stratification can influence cyanobacterial bloom dynamics [6] as well as water treatment. Ultimately, the lake is influenced by both rapid short-term change (affecting thermal dynamics, blooms, biogeochemistry and also influencing toxins and water treatability) and longer-term patterns of change associated with climate, water management, and likely reservoir aging and watershed loading. Jointly these drivers contribute to changes in source water quality that influence treatment [3,6,12,13,17] and have motivated major recent investments in upgraded infrastructure [18].

The design of the buoy and sensor system (named “big buoy”) had the joint goal of supporting science to understand the limnology of polymictic lakes (including drivers of rapid change) and supporting work of the water treatment plant. As such, the monitoring station was near one of the water treatment intakes, located in the downstream portion of the reservoir (see [6]). The monitoring system was designed to provide hourly data transmission via cellular telemetry allowing plant operators to see and adapt to changes in physical, chemical and biotic attributes of incoming water during the open water (ice-free) season. It was also designed to allow adaptive sampling to target specific conditions (e.g., severe blooms) and quickly identify operational problems.

4.2. Station setup and design

Continuous water quality and meteorological monitoring is conducted using a multiparameter sensor array deployed on a stationary buoy platform (CB-800; Coastal Data buoy;

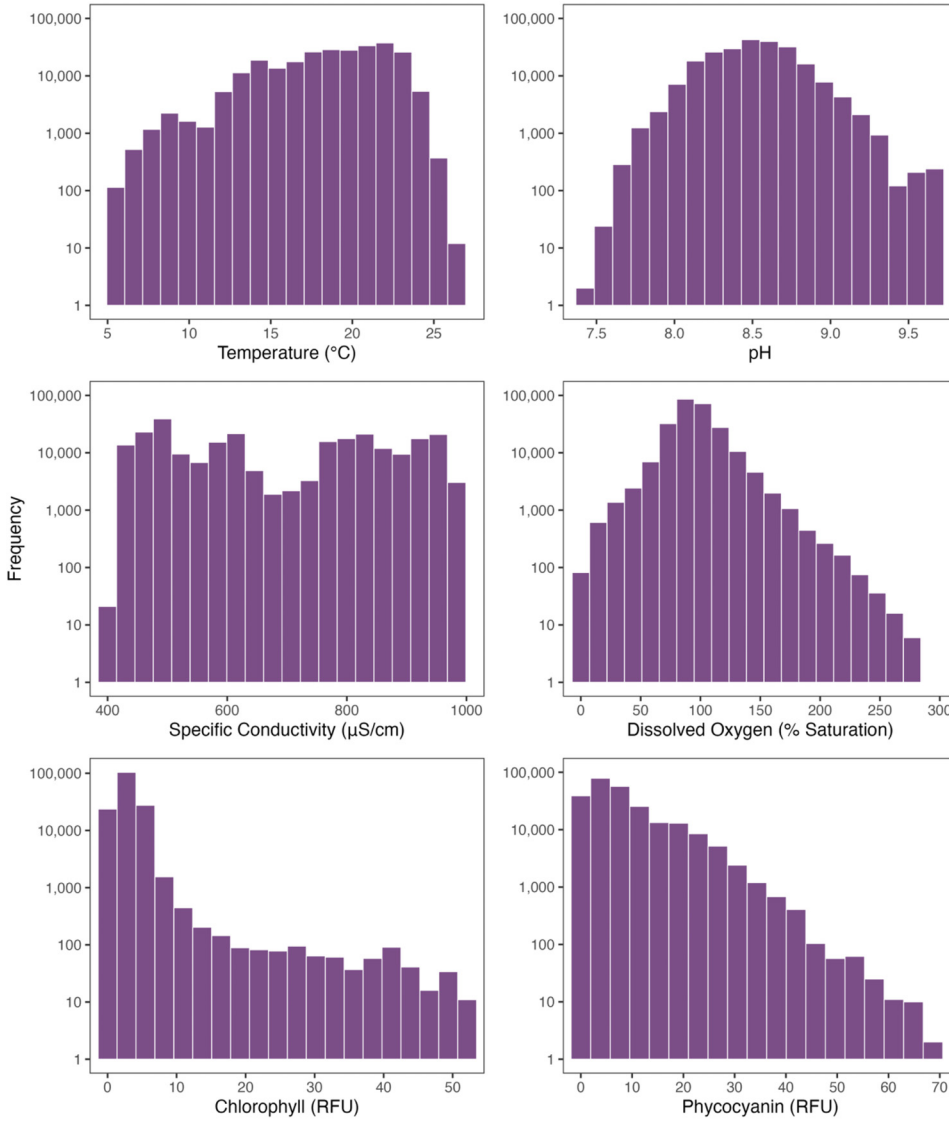

Fig. 1. Frequency distributions for select water quality variables (temperature, pH, oxygen % saturation, specific conductivity and pigments for the shallow sondes) across all years of data collection. Note log scale on y-axis.

NexSens) (Fig. 2) in Buffalo Pound Lake, Saskatchewan, Canada. Annual deployment occurs during the ice-free period, with buoy installation typically conducted between May and June and retrieval between September and October (Table 5). The sensor suite used has remained largely consistent through time, with minor changes (Tables 6 & 7). The system is solar powered, with cellular telemetry. It is moored via a two-point anchor system (using 70-pound pyramid anchors) which minimizes wind-related changes in direction and movement in this sometimes high turbulence environment.

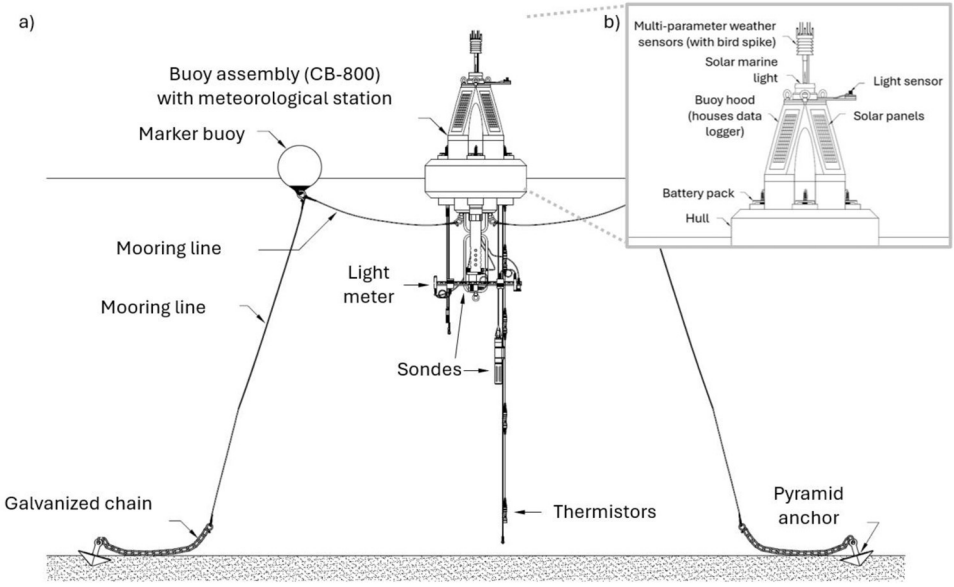

Fig. 2. Schematic of buoy setup: a) overall system b) details of above-water instrumentation. Note that depths of instrumentation are not to scale (see text for details). Sondes house multiple sensors (i.e., temperature, pH, dissolved oxygen, specific conductivity, turbidity chlorophyll, phycocyanin; see text for details). Sondes and underwater light meters have wipers (see text and tables for details on which sensors are wiped). Line diagrams are modified from figures developed by Nexsens technology, Inc and reproduced with permission.

Table 5
Annual deployment dates, sampling intervals, and coordinates for Buffalo Pound Lake monitoring buoy (2014–2021).

Deployment (YYYY-MM-DD)	Removal (YYYY-MM-DD)	Sampling Interval (min)	Latitude	Longitude
2014–06–11	2014–09–04	15	50.586310	−105.384921
2015–05–14	2015–10–06	10	50.585723	−105.387343
2016–05–18	2016–09–28	10	50.585662	−105.384507
2017–05–09	2017–10–11	10	50.585807	−105.385008
2018–05–15	2018–10–04	10	50.585620	−105.385010
2019–05–29	2019–10–03	10	50.585792	−105.383540
2020–06–05	2020–10–01	10	50.585611	−105.384320
2021–05–31	2021–10–07	10	50.586120	−105.384190

4.3. Meteorological sensors/ weather station

Meteorological data was collected using a Vaisala weather station mounted on the buoy platform, 1.66 m above the water surface. Measurements include barometric pressure, air temperature, relative humidity, wind direction, wind speed and daily rain. Incident photosynthetically active radiation (PAR) was measured using a LI-COR sensor mounted 1.06 m above the water surface (Table 6).

4.4. Water quality monitoring, temperature profiles and light measurements

Water quality data were collected using an array of multi-parameter sondes and thermistors (NexSens T-Node FR titanium nodes at multiple depths). Sondes were deployed at two depths (Table 7), measuring temperature, specific conductivity, pH, turbidity, chlorophyll, phycocyanin,

Table 6
Meteorological sensor and photosynthetically active radiation (PAR) specifications and deployment details for above-surface measurements at Buffalo Pound Lake monitoring station.

Variable	Column name	Sensor	Sensor Model	Sensor Type	Depth (m)	Accuracy	Operating Range	Years active
Battery	Bat_V	iSIC Battery	Lithium	Solar Powered	N/A	N/A	0 – 15 V	2014 – 2021
Barometric pressure	BarometricPress_kPa	Vaisala	WXT536	Capacitive silicon BAROCAP®	1.66	± 0.5 kPa (0 – 30 °C)	60 – 110 kPa	2014 – 2021
Relative humidity	RelativeHum_%	Vaisala	WXT536	Capacitive thin film polymer HUMICAP®	1.66	± 3 %	0 – 100 % RH	2014 – 2021
Wind direction	WindDir_Degree	Vaisala	WXT536	Ultrasonic transducers (3)	1.66	± 3 °C	0 – 360°	2014 – 2021
Wind speed	WindSp_km/h	Vaisala	WXT536	Ultrasonic transducers (3)	1.66	± 3 %	0 – 216 km h^{-1}	2014 – 2021
Daily rain	DailyRain_mm	Vaisala	WXT536	Piezoelectrical sensor plate (RAINCAP®)	1.66	± 5 %	0 – 200 mm h^{-1}	2014 – 2021
Air temperature	AirTemp_C	Vaisala	WXT536	Capacitive ceramic THERMOCAP®	1.66	± 3 °C	2 – 60 °C	2014 – 2021
PAR air	PARAir_umol/s/m^2	LI-COR	LI-190	Quantum	1.06	± 5 %	2 – 2,000 µmol s^{-1} m^{-2}	2014 – 2021

Table 7
Multi-depth sensor specifications and deployment details for water quality, photosynthetically active radiation (PAR), and temperature profile measurements at Buffalo Pound Lake monitoring station. (Note that multiple measurements are sometimes reported for single sensors; e.g. Optical DO as % saturation and mg/L). Wiped sensors are noted under sensor type.

Variable	Column name	Sensor	Sensor Model	Sensor Type	Depth (m)	Accuracy	Operating Range	Years active
PAR water 1	PARW1_umol/s/m^2	LI-COR	LI-192	Underwater (wiped)	0.62	± 5 %	2 – 2,000 µmol s^{-1} m^{-2}	2014 – 2021
PAR water 2	PARW2_umol/s/m^2	LI-COR	LI-192	Underwater (wiped)	0.78	± 5 %	2 – 2,000 µmol s^{-1} m^{-2}	2014 – 2021
Temperature	TempShallow_C	YSI Sonde 6600	6560	Titanium	0.82	± 0.15 °C	−5 – 50 °C	2014 – 2021
pH	pHShallow	YSI Sonde 6600	6569 or 6561	Unguarded glass combination electrode	0.82	± 0.2	0 – 14 pH units	2014 – 2021
pH mV	pHmVShallow_mV	YSI Sonde 6600	6569 or 6561	See above[1]	0.82	± 20 mV	N/A	2014 – 2021
Optical Dissolved Oxygen	ODOSatShallow_%	YSI Sonde 6600	6150+ROX	Optical, luminescence (wiped)	0.82	± 1 % (0 – 200 %) ± 15 % (200 – 500 %)	0 – 500 % saturation	2014 – 2021
Optical Dissolved Oxygen	ODOShallow_mg/L	YSI Sonde 6600	6150+ROX	Optical, luminescence (wiped)	0.82	± 0.1 mg/L (0 – 20 mg/L) ± 15 % (20 – 50 mg/L)	0 – 50 mg L^{-1}	2014 – 2021
Specific conductivity	SpCondShallow_uS/cm	YSI Sonde 6600	6560	4-electrode cell	0.8	± 0.5 %	0 – 100,000 µS cm^{-1}	2014 – 2021
Turbidity	TurbShallow_NTU+	YSI Sonde 6600	6136	Optical, 90° scatter (wiped)	0.82	0.3 NTU or ± 2 %	0 – 1,000 NTU[3]	2014 – 2021
Chlorophyll	ChlRFUShallow_RFU	YSI Sonde 6600	6025	Optical, fluorescence (wiped)	0.82	Not Specified	0 – 100 RFU[2]	2014 – 2021
Phycocyanin blue-green algae	BGPCShallowRFU_RFU	YSI Sonde 6600	6131	Optical, fluorescence (wiped)	0.82	Not Specified	0 – 100 RFU[2]	2014 – 2021
Temperature	TempDeep_C	YSI Sonde 6920	6560	Titanium Thermistor	2.83	± 0.15 °C	−5 – 50 °C	2014 – 2019
pH	pHDeep	YSI Sonde 6920	6565 or 6561	Glass combination electrode	2.83	± 0.2	0 – 14 pH units	2014 – 2019
pH mV	pHmVDeep_mV	YSI Sonde 6920	6565 or 6561	See above[1]	2.83	± 20 mV	N/A	2014 – 2019
Optical Dissolved Oxygen	ODOSatDeep_%	YSI Sonde 6920	6150+ROX	Optical, luminescence (wiped)	2.83	± 1 % (0 – 200 %) ± 15 % (200 – 500 %)	0 – 500 % saturation	2014 – 2019
Optical Dissolved Oxygen	ODODeep_mg/L	YSI Sonde 6920	6150+ROX	Optical, luminescence (wiped)	2.83	± 0.1 mg/L (0 – 20 mg/L) ± 15 % (20 – 50 mg/L)	0 – 50 mg L^{-1}	2014 – 2019

(*continued on next page*)

Table 7 (*continued*)

Variable	Column name	Sensor	Sensor Model	Sensor Type	Depth (m)	Accuracy	Operating Range	Years active
Specific conductivity	SpCondDeep_uS/cm	YSI Sonde 6920	6560	4-electrode cell	2.8	± 0.5 %	0 – 100,000 µS cm^{-1}	2014 – 2019
Phycocyanin blue-green algae	BGPCDeepRFU_RFU	YSI Sonde 6920	6131	Optical, fluorescence (wiped)	2.83	Not Specified	0 – 100 RFU[2]	2014 – 2019
Temperature	TempDeep_C	EXO2 Sonde	599872	Titanium Thermistor (wiped)	2.83	± 0.2 °C	−5 – 50 °C	2020 – 2021
pH	pHDeep	EXO2 Sonde	577602	Unguarded glass combination electrode (wiped)	2.83	± 0.1	0 – 14 pH units	2020 – 2021
pH mV	pHmVDeep_mV	EXO2 Sonde	577602	See above[1]	2.83	± 20 mV	N/A	2020 – 2021
Optical Dissolved Oxygen	ODOSatDeep_%	EXO2 Sonde	599100–01	Optical, luminescence (wiped)	2.83	± 1 % (0 – 200 %) ± 5 % (200 – 500 %)	0 – 500 % saturation	2020 – 2021
Optical Dissolved Oxygen	ODODeep_mg/L	EXO2 Sonde	599100–01	Optical, luminescence (wiped)	2.83	± 0.1 mg/L (0 – 20 mg/L) ± 5 % (20 – 50 mg/L)	0 – 50 mg L^{-1}	2020 – 2021
Specific conductivity	SpCondDeep_uS-cm	EXO2 Sonde	599872	Titanium Thermistor (wiped)	2.8	± 1 %	0 – 100,000 µS cm^{-1}	2020 – 2021
Chlorophyll	ChlorophyllRFUDeep_RFU	EXO2 Sonde	599102–01	Optical dual-channel digital smart sensor (wiped)	2.83	± 0.01 RFU[2]	0 – 100 RFU[2]	2020 – 2021
TAL Phycocyanin	BGPCDeepRFU_RFU	EXO2 Sonde	599102–01	Optical dual-channel digital smart sensor (wiped)	2.83	± 0.01 RFU[2]	0 – 100 RFU[2]	2020 – 2021
Temperature	Temp00_C	Nexsens	T-Node FR	Titanium	0.54	± 0.075 °C	0 – 45 °C	2015 – 2021
Temperature	Temp01_C	Nexsens	T-Node FR	Titanium	1.06	± 0.075 °C	0 – 45 °C	2014 – 2021
Temperature	Temp02_C	Nexsens	T-Node FR	Titanium	1.56	± 0.075 °C	0 – 45 °C	2014 – 2021
Temperature	Temp03_C	Nexsens	T-Node FR	Titanium	2.56	± 0.075 °C	0 – 45 °C	2014 – 2021
Temperature	Temp04_C	Nexsens	T-Node FR	Titanium	3.56	± 0.075 °C	0 – 45 °C	2015 – 2021

[1] pH mV is included as a diagnostic for pH measurements from the associated sensor.
[2] Relative fluorescence units.
[3] Turbidity data are reported in NTU, consistent with the practice of data users, including the water treatment plant, as well as long-term datasets and published manuscripts. Based on the International Standards Organization (ISO) method 7027, the unit's name is more correctly Formazin Nephelometric Units (FNU). Measured values are unchanged based on choice of naming convention.

and dissolved oxygen, with specific parameters measured dependent on the sonde model installed and depth (Table 7). Sondes were equipped with YSI self-cleaning wipers (see Table 7 for details, including which sensors were wiped). Photosynthetically active radiation (PAR) in the water column was measured in the water column at 0.62 and 0.78 m below the lake surface (LI-COR LI-192 equipped with Zebra-Tech H-SC-001–030 wipers). Both chlorophyll and phycocyanin are reported in relative fluorescence units (RFU) rather than as concentrations.

4.5. System maintenance

The systems were maintained through the open water season. Although most YSI sensors are wiped (Table 7), fouling could still occur, as a result, visits included an inspection for fouling, and removal of any visible fouling. Maintenance intervals were varied based on fouling risk and spanned a target of approximately weekly maintenance during the period of peak water temperature and blooms (i.e., late June through early September) and declining to approximately every two weeks in other periods (spring and autumn). During maintenance, in addition to assessment of fouling and removal of any materials, the dissolved oxygen, pH, and specific conductivity sensors underwent calibration according to manufacturer specifications. Specifically, pH underwent a three-point calibration (pH 4, 7 and 10 certified buffers; Fisher Scientific), oxygen was calibrated with air-saturated water, and the conductivity sensor was calibrated with a certified standard (1413 µS). Other sensors (i.e., chlorophyll, phycocyanin and turbidity) were found to show minimal drift, hence were typically calibrated only at the start and end of the field seasons. Both the turbidity and pigment sensors were calibrated using a two-point calibration, including high purity water (milliQ®) and commercially available standards. The turbidity sensor calibration used a YSI polymer-based standard (YSI 6073 G), while rhodamine solutions are used to calibrate pigment sensors (both in accordance with sensor manufacturer instructions). During each maintenance visit, pre- and post-maintenance values were used to assess biofouling (and instrumental drift for field-calibrated parameters). A flag of B7 was used to identify likely periods where fouling could impact data quality. Because the onset of fouling is difficult to diagnose, the impacted period was defined as the entire duration between the last site maintenance visit and the subsequent visit where the issue was identified. Drift was consistently minimal for all sensors (typically within the range of expected sensor error; Table 7).

4.6. Data handling

As noted, we used a dual approach of reviewing field-flagged data (i.e., notes of damage, fouling, periods of maintenance visits) and performing automated screening (for missing data, data outside operating ranges or historic ranges, and erratic changes). Any field flagged data or automatically flagged data were then manually reviewed. Where this manual review indicated that data were erroneous, they were removed and the resulting data gap was flagged with the appropriate label (Table 2; flag C). This included faulty data associated with sensor failure or electrical problems. For example, measurements of barometric pressure or conductivity of zero were omitted. A number of measurements of pH <5 were flagged as locally unrealistic and later diagnosed as associated with sensor failure, hence data were omitted. Finally, automated screening identified erratic changes such as an approximately 20 °C increase in measured water column temperatures in a 4 h period which was associated with failure of a thermistor. The affected periods were flagged with erroneous data removed.

Light data from nighttime periods were often below the operating range (Tables 6 & 7), and in some cases, light sensors reported slightly negative PAR values (e.g., -2 μmol s^{-1} m^{-2}). Any values for PAR below the sensor operating range (which has a minimum of 2 μmol s^{-1} m^{-2}) were replaced with values of 0 μmol s^{-1} m^{-2} and these data were flagged (flag A2; Table 2). Other data such as those showing higher PAR at the lower water column depth are flagged for

Table 8
Example excerpt of the Buffalo Pound Lake buoy dataset showing timestamped measurements with no associated quality flag.

DateTime	Bat_V	BarometricPress_kPa	BarometricPress_kPa_Flag
2014–06–11 22:15	12.5	95.8	
2014–06–11 22:30	12.6	95.8	
2014–06–11 22:45	12.6	95.8	

Table 9
Example excerpt of the Buffalo Pound Lake buoy dataset showing missing measurements at the given timestamp and the associated removal flag.

DateTime	Bat_V	BarometricPress_kPa	BarometricPress_kPa_Flag
2014–08–22 13:45	12.3	NA	C
2014–08–22 14:00	12.1	NA	C
2014–08–22 14:15	12	NA	C

user inspection (flag A3; Table 2) but are not changed. Finally, all water column PAR data for the year 2015 are flagged, because the deployment orientation could lead to shading of one sensor, hence users are directed to use the higher value of the two sensors (flag A4; Table 2 or see data_flags.csv).

Calibration errors in specific conductivity measurements were identified in 2017 in both shallow (0.8 m) and deep (2.8 m) sensors and in 2018 in the deep sensor alone. This resulted in an offset in the data. The data were adjusted by working backward from new calibration standards, then data were compared to manual measurements (handheld YSI) and chemical data analysis from the Buffalo Pound Water Treatment Plant. The applied correction reduced the impact of the error, although effects remain, and affected data are noted in a data flag (flag A6).

Oftentimes, automatically flagged data based on historic operating ranges represented true, but extreme conditions. For example, as part of the automated screening, outliers were identified, but often represented environmental events such as algal blooms, high wind conditions or sudden rainfall events. Following manual inspection of all automatically flagged data, we removed flags where data were deemed acceptable (e.g., where automated flags were associated with these environmental events). This included both rapid change (e.g., storm events or bloom conditions), and excursions from historic operating ranges.

The final data are consistent with expected values for the regional conditions, seasonal patterns, and environmental variability typical of Buffalo Pound Lake. The final data (following this review process) include dates and times for the full time series for each deployment period with the exception of periods of maintenance. These maintenance periods are known to lead to erroneous values, hence are omitted from the dataset. Within this remaining time series, data points were either determined to be acceptable with all flags removed (example in Table 8), classified as faulty data with explanatory removal flags (flag C, example in Table 9), or assigned quality flags denoting data that are likely accurate but represent unusual conditions, or have cautions associated with their use (see data_flags.csv for complete data quality flag definitions or Table 2). Missing measurements are also flagged. Through the time series, almost all parameters have >120,000 total measurements reported (Table 4), typically with >15,000 measurements reported per sensor in a given year (Table 3).

Limitations

Sensor-based data have inherent limitations associated with sensor specifications (sensitivity, measurement ranges) as well as issues of fouling, maintenance and failure. Data have been screened to remove known erroneous data (e.g., sensor failure and maintenance periods). The re-

maining data are flagged to indicate areas data users should further interrogate data quality for their purposes. Notable among them is limitations in the 2015 PAR data associated with periodic shading due to sensor orientation in that deployment year, and conductivity data in 2017 and 2018. There are data gaps. For example, the deep pH sensor failed in 2021 and due to difficulty diagnosing the date of failure, all 2021 deep pH data were removed. Pigment sensors (chlorophyll and phycocyanin) are reported in relative units, limiting the types of analyses that can be done without inclusion of lab-based measurements of concentration. Minor changes in the sensor arrays did occur (e.g. at the end of their usable life) (see Tables 6 & 7). Users should review the metadata and data flags to ensure suitability for their intended uses and are encouraged to reach out to authors to contextualize the data with insights on the function of this ecosystem and sensors.

Ethics Statement

This work did not include human or animal subjects, or use of social media, and follows the ethical requirements for publication in Data in Brief.

CRediT author statement

Helen Baulch: Conceptualization, methodology, resources, writing-original draft, data curation, writing-review & editing, supervision, project administration, funding acquisition. **Jay Bauer:** Methodology, data curation, writing-review & editing. **Lisa Boyer:** Methodology, Data curation, Writing-original draft, writing-review & editing, visualization. **Katy Nugent:** Methodology, Validation, data curation, writing-original draft, writing-review & editing. **Kristin Painter:** Writing – original draft, writing-review & editing, data curation. **Jason J. Venkiteswaran:** Data curation, methodology, funding acquisition, project administration, software, supervision, writing-review & editing. **Lana Vuleta:** software, data curation, methodology, writing-review & editing. **Colin Whitfield:** software, data curation, funding acquisition, methodology, project administration, resources, supervision, writing-review & editing

Data Availability

A long-term water quality and meteorological data set (2014 – 2021) of a eutrophic prairie lake: Buffalo Pound Lake, Saskatchewan, Canada (Original data) (FRDR).

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Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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